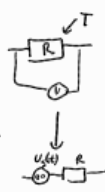


$\langle I(t) \rangle = 0$
 $\langle I^2 \rangle \neq 0$
 $U_j(t) = I(t)R$
 $\langle U_j(t) \rangle = 0$
 $\langle U_j^2 \rangle \neq 0$



Ocenja $U_j(t)$?

$\langle U_j(t) U_j(t+\tau) \rangle = q \delta(\tau)$ nekorelirana šuma



$U_j - IR - \frac{q}{C} = 0$
 $\frac{dq}{dt} = I = (C U_c)' = C \dot{U}_c$
 $U_j - RC \dot{U}_c - U_c = 0$
 $\dot{U}_c = -\frac{1}{RC} U_c + \frac{1}{RC} U_j$

Kalmanov filter:

$X = U_c$
 $\dot{U}_c = A U_c + \Gamma w$
 $A = -\frac{1}{RC}$
 $\Gamma = \frac{1}{RC}$
 $w = U_j(t)$

Kovariacija

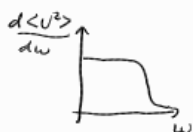
$P = \langle (U_c - \langle U_c \rangle)^2 \rangle = \langle U_c^2 \rangle$
 $\dot{P} = -\frac{2}{RC} P + \frac{1}{RC} q$
 stac. rež. (termodinamsko ravnovesje)
 $q = 2\tau P_{ss}$

Ekvipart. izrek

$\langle W_c \rangle = \frac{1}{2} \langle U_c^2 \rangle = \frac{1}{2} k_B T$
 $\rightarrow \langle U_c^2 \rangle = \frac{k_B T}{C} \sim 60 \mu V$
 $q = 2\tau \frac{k_B T}{C} = 2 k_B R T$

Spektralna gostota termičnega šuma

$\langle U_j(t) U_j(t+\tau) \rangle = \delta(\tau) 2 k_B R T$



$P = \frac{U^2}{R}$
 $\langle P \rangle = \frac{\langle U^2 \rangle}{R}$
 $P = \int_{-\infty}^{\infty} \frac{dP}{d\omega} d\omega \rightarrow \infty!$

$\frac{d\langle P \rangle}{d\omega} = \frac{d\langle U^2 \rangle}{d\omega} \frac{1}{R}$

Wiener-Hinčinov izrek

Spektralna gostota šuma je enaka

Fourierovi transformaciji njegoše avtokorelacijske funkcije

$\langle C(\tau) \rangle = \int_{-\infty}^{\infty} U^*(t) U(t+\tau) dt$ avtokorelacijska funkcija

$U(t) = \int_{-\infty}^{\infty} U_\nu e^{-2\pi i \nu t} d\nu$

$U^*(t) = \int_{-\infty}^{\infty} U_\nu^* e^{2\pi i \nu t} d\nu$

$\langle C(\tau) \rangle = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} U_\nu^* e^{2\pi i \nu t} d\nu \int_{-\infty}^{\infty} U_{\nu'} e^{-2\pi i \nu' (t+\tau)} d\nu' dt =$

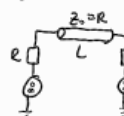
$= \iint U_\nu^* U_{\nu'} \underbrace{\int_{-\infty}^{\infty} e^{2\pi i (\nu - \nu') t} dt}_{\delta(\nu - \nu')} e^{-2\pi i \nu' \tau} d\nu d\nu' =$

$= \int_{-\infty}^{\infty} |U_\nu|^2 e^{-2\pi i \nu \tau} d\nu$

$|U_\nu|^2 = \frac{1}{\pi} \int_{-\infty}^{\infty} e^{-2\pi i \nu \tau} \langle C(\tau) \rangle d\tau$
 $\downarrow$
 $\delta(\tau) 2 k_B R T$

$\frac{d\langle U^2 \rangle}{d\omega} = \frac{1}{\pi} 2 k_B R T$

Nyquistova izpoljava realne spektralne gostote termičnega šuma



$\lambda = \frac{c}{\nu} = \frac{2\pi}{\omega} c$

$L = h \frac{\lambda}{c}$ pogoj za stojice valovanje v kablu

$L = h \frac{\pi}{\omega} c$

$h = \frac{L}{\pi c} \omega \rightarrow \frac{d h}{d \omega} = \frac{L}{\pi c}$

Energija stojčnega valovanja

$\xi(\omega_n) = h \omega_n f(\omega_n) = h \omega_n \left(\frac{1}{e^{h \omega_n / k_B T} - 1} \right)$

$dP = \frac{1}{2} \left(\frac{K d\omega}{\pi c} \right) \xi(\omega_n) \frac{c}{L} = \frac{1}{2\pi} h \omega_n \frac{1}{e^{h \omega_n / k_B T} - 1} d\omega$

$I = \frac{U}{2R}$

$\langle I^2 \rangle = \frac{\langle U^2 \rangle}{4R^2}$

$d\langle P \rangle = R d\langle I^2 \rangle = \frac{d\langle U^2 \rangle}{4R} = \frac{1}{2\pi} h \omega \frac{1}{e^{h \omega / k_B T} - 1} d\omega$

$\frac{d\langle U^2 \rangle}{d\omega} = \begin{cases} \frac{2 k_B R T}{\pi} & ; \frac{h \omega}{k_B T} \ll 1 \\ 0 & ; \frac{h \omega}{k_B T} \gg 1 \end{cases}$ CUTOFF

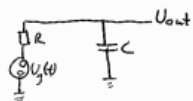
$\frac{h \omega}{k_B T} \ll 1 \rightarrow \frac{1}{e^{h \omega / k_B T} - 1} \sim \frac{k_B T}{h \omega}$

CUTOFF se zgodi pri

$k_B T / h \omega \approx 1 \rightarrow \omega \sim 10^{13} \text{ Hz}$

Širjenje termičnega šuma skozi linearna vezja

np. RC člen



$$U_{out} = H(s) U_g(s)$$

$$U_{out}^2(i\omega) = |H(i\omega)|^2 U_g^2(i\omega)$$

$$\frac{d\langle U_{out}^2 \rangle}{d\omega} = |H(i\omega)|^2 \frac{d\langle U_g^2(i\omega) \rangle}{d\omega}$$

$$H(s) = \frac{1}{1+RCs}$$

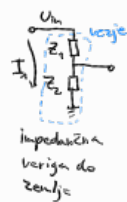
$$H(i\omega) = \frac{1}{1+i\omega RC}$$

$$\frac{d\langle U_{out}^2 \rangle}{d\omega} = \frac{1}{1+\omega^2 R^2 C^2} \frac{2k_B T}{\pi}$$

izhodna impedanca vezja

$$\frac{d\langle U_{out}^2 \rangle}{d\omega} = \frac{2k_B T}{\pi} \text{Re}(Z_o)$$

Nyquistov izrek



1) open circuit:

$$U_{in} = Z I_1 = (Z_1 + Z_2) I_1$$

Z_{in}
vhodna
impedanca
vezja

$$U_{out} = \frac{Z_2 U_{in}}{Z_1 + Z_2}$$

$$Z_{in} = \sum_i Z_i$$

2) out kratko skleneno
(short circuit)



$$I_{sc} = \frac{U_{in}}{Z_1}$$

$$Z_{out} I_{sc} = U_{out}$$

$$U_{out} = \frac{Z_2}{Z_1 + Z_2} U_{in} = \frac{Z_2}{Z_1 + Z_2} Z_1 I_{sc}$$

$$Z_{out} = \frac{Z_1 Z_2}{Z_1 + Z_2} = \left(\frac{1}{Z_1} + \frac{1}{Z_2} \right)^{-1}$$

$$Z_{out}^{-1} = \sum_i Z_i^{-1}$$